

NPTEL • EXAM QUICK REVISION

Cloud Computing

MCQ & MSQ Question Bank

75 MCQs • 29 MSQs • Answers highlighted

About this bank (honest note)

- This PDF is **only** MCQs (single correct) and MSQs (multiple correct), as requested.
- Every question is grounded **only** in the provided Cloud Computing Lecture 9, 10 & 11 transcripts — nothing invented.
- Topics: Cloud/Fog/Edge, resource management & service placement, federation, VM migration, containers/Docker/Kubernetes, dew computing, serverless, and sustainable cloud computing.
- **MCQ** = pick one. **MSQ** = select all that apply. Correct answers are shown in the green box under each question.

Part A — Cloud, Fog & Edge Fundamentals

MCQ single correct answer **MSQ** multiple correct answers — select all. Correct option(s) highlighted below each question.

MCQ Q1. The term 'fog computing' was originally coined by which company?

- A. Google
- B. Amazon
- C. Cisco Systems
- D. Microsoft

Answer: C. Cisco Systems

MCQ Q2. In the cloud–fog–dew analogy, fog computing is best described as:

- A. Far above in the sky
- B. Concentrated at the network edge, closer to devices
- C. Only inside data centers
- D. A replacement for the cloud

Answer: B. Concentrated at the network edge, closer to devices

MCQ Q3. Fog computing is best characterised as:

- A. A complete replacement for cloud computing
- B. A complementary/supportive technology to the cloud
- C. A type of database service
- D. A storage-only service

Answer: B. A complementary/supportive technology to the cloud

MCQ Q4. Which service model gives the user only the basic computing infrastructure (compute, storage) to build on?

- A. SaaS
- B. PaaS
- C. IaaS
- D. DaaS

Answer: C. IaaS

MCQ Q5. In which model does the user 'not care about the infrastructure' and directly uses the working software?

- A. IaaS
- B. PaaS
- C. SaaS
- D. Fog

Answer: C. SaaS

MCQ Q6. Compared to cloud, fog computing typically offers:

- A. Higher latency
- B. Lower latency
- C. No network access
- D. Larger centralized storage

Answer: B. Lower latency

MCQ Q7. The in-house open-source cloud infrastructure of IIT Kharagpur mentioned in the lectures is called:

- A. OpenStack
- B. Meghamala
- C. CloudSim
- D. Megha-fog

Answer: B. Meghamala

MCQ Q8. Which simulator was developed by a combined team of IIT Kharagpur and University of Melbourne?

- A. CloudSim
- B. iFogSim
- C. GridSim
- D. NS-3

Answer: B. iFogSim

MCQ Q9. CloudSim originates from which institution?

- A. IIT Kharagpur
- B. University of Melbourne
- C. Stanford
- D. MIT

Answer: B. University of Melbourne

MCQ Q10. The vision of fog computing is to:

- A. Push all data to the cloud
- B. Run applications directly at the network edge to reduce overall traffic
- C. Eliminate sensors
- D. Store data only on mobile devices

Answer: B. Run applications directly at the network edge to reduce overall traffic

MCQ Q11. A key limitation of edge/IoT end devices is:

- A. Excess storage
- B. Limited processing power, storage and power/battery
- C. Global visibility
- D. Unlimited bandwidth

Answer: B. Limited processing power, storage and power/battery

MCQ Q12. In the health case study, which module was placed on the cloud because the cloud has the global view?

- A. Client module
- B. Data filter module
- C. Conformity/confirmation module
- D. Display module

Answer: C. Conformity/confirmation module

MSQ Q13. Select ALL essential characteristics of cloud computing mentioned:

- A. On-demand self-service
- B. Broad network access
- C. Resource pooling
- D. Permanent offline mode
- E. Rapid elasticity
- F. Measured service

Answers: A, B, C, E, F — On-demand self-service; Broad network access; Resource pooling; Rapid elasticity; Measured service

MSQ Q14. Which of the following are TRUE advantages of the cloud–fog–edge paradigm over cloud-only?

- A. Reduced latency
- B. Reduced network traffic to the cloud
- C. Better location awareness at the fog
- D. Elimination of the cloud entirely
- E. Faster real-time response

Answers: A, B, C, E — Reduced latency; Reduced network traffic to the cloud; Better location awareness at the fog; Faster real-time response

MSQ Q15. Which devices can act as fog nodes (given adequate resources)?

- A. Network routers
- B. Wi-Fi gateways
- C. Cellular base stations
- D. The generating power station
- E. Set-top boxes / small servers

Answers: A, B, C, E — Network routers; Wi-Fi gateways; Cellular base stations; Set-top boxes / small servers

Part B — Resource Management & Service Placement

MCQ Q16. In the resource-management taxonomy, the architecture aspect is classified by data flow, tenancy and:

- A. Cooling
- B. Control
- C. Cost
- D. Colour

Answer: B. Control

MCQ Q17. Which is NOT one of the three infrastructure components for fog/edge resource management?

- A. Hardware
- B. System software
- C. Middleware
- D. Firmware compiler

Answer: D. Firmware compiler

MCQ Q18. Middleware in a fog/edge system primarily provides:

- A. CPU manufacturing
- B. Coordination, orchestration, monitoring and communication services
- C. Physical cooling
- D. Raw electricity

Answer: B. Coordination, orchestration, monitoring and communication services

MCQ Q19. Which algorithm component identifies edge resources available for distributed computing?

- A. Benchmarking
- B. Discovery
- C. Load balancing
- D. Placement

Answer: B. Discovery

MCQ Q20. Capturing the performance (CPU, storage, networking) of resources for decision-making is called:

- A. Discovery
- B. Benchmarking
- C. Placement
- D. Tenancy

Answer: B. Benchmarking

MCQ Q21. In offloading, the user-device-to-edge direction commonly uses:

- A. Server replication only
- B. Application partitioning and caching
- C. VM migration only
- D. Cold migration

Answer: B. Application partitioning and caching

MCQ Q22. Cloud-to-edge offloading commonly uses:

- A. Application partitioning at the user
- B. Server offloading (replication/partitioning) and caching
- C. Only DVFS
- D. Only pruning

Answer: B. Server offloading (replication/partitioning) and caching

MCQ Q23. A requirement that a service must execute within a specific location is an example of a:

- A. Resource-level constraint
- B. Network-level constraint
- C. Application-level constraint
- D. Cooling constraint

Answer: C. Application-level constraint

MCQ Q24. Software Defined Networking (SDN) is associated with which kind of virtualization?

- A. System virtualization
- B. Network virtualization
- C. Storage virtualization
- D. Memory virtualization

Answer: B. Network virtualization

MSQ Q25. Select ALL components of the resource-management ALGORITHM group:

- A. Discovery
- B. Benchmarking
- C. Load balancing
- D. Cooling
- E. Placement

Answers: A, B, C, E — Discovery; Benchmarking; Load balancing; Placement

MSQ Q26. Which are recognised control models for managing resources?

- A. Centralized
- B. Distributed
- C. Hierarchical
- D. Randomized-only

Answers: A, B, C — Centralized; Distributed; Hierarchical

MSQ Q27. Service-placement constraints include which of the following?

- A. Resource-level (CPU/RAM/storage limits)
- B. Network-level (latency/bandwidth)
- C. Application-level (locality, delay-sensitivity)
- D. Marketing-level

Answers: A, B, C — Resource-level (CPU/RAM/storage limits); Network-level (latency/bandwidth); Application-level (locality, delay-sensitivity)

MSQ Q28. Under 'infrastructure', which are covered?

- A. Hardware (compute + network devices)
- B. System software (OS, virtualization)
- C. Middleware
- D. Application business logic itself

Answers: A, B, C — Hardware (compute + network devices); System software (OS, virtualization); Middleware

Part C — Cloud Federation

MCQ Q29. Cloud federation is best defined as:

- A. A single private cloud
- B. A union of multiple internal/external clouds cooperating to meet a business need
- C. A container runtime
- D. A load balancer only

Answer: B. A union of multiple internal/external clouds cooperating to meet a business need

MCQ Q30. Which federation architecture combines an on-premise private cloud with public cloud for peak demand?

- A. Broker
- B. Aggregation
- C. Cloud bursting / hybrid
- D. Multi-tier

Answer: C. Cloud bursting / hybrid

MCQ Q31. The cloud-bursting/hybrid architecture is typically:

- A. Tightly coupled
- B. Loosely coupled
- C. Partially coupled
- D. Uncoupled

Answer: B. Loosely coupled

MCQ Q32. Which architecture uses an intermediary that selects/deploys resources across several public clouds?

- A. Broker
- B. Multi-tier
- C. Bursting
- D. Aggregation

Answer: A. Broker

MCQ Q33. The multi-tier federation architecture is:

- A. Loosely coupled and flat
- B. Tightly coupled and hierarchical
- C. Partially coupled and broker-based
- D. Uncoupled

Answer: B. Tightly coupled and hierarchical

MCQ Q34. Aggregation architecture is usually described as:

- A. Loosely coupled
- B. Partially coupled
- C. Tightly coupled
- D. Voluntary only

Answer: B. Partially coupled

MCQ Q35. In a loosely coupled federation, a CSP has:

- A. Full control over remote VM placement
- B. Little or no control over remote resources
- C. Complete monitoring of remote hosts
- D. Cross-site VM migration by default

Answer: B. Little or no control over remote resources

MCQ Q36. Advanced features such as cross-site networks and cross-site VM migration are possible mainly in:

- A. Loosely coupled federation
- B. Partially coupled federation
- C. Tightly coupled federation
- D. No federation

Answer: C. Tightly coupled federation

MSQ Q37. Select ALL cloud-federation architectures discussed:

- A. Cloud bursting / hybrid
- B. Broker
- C. Aggregation
- D. Multi-tier
- E. Peer-review

Answers: A, B, C, D — Cloud bursting / hybrid; Broker; Aggregation; Multi-tier

MSQ Q38. Select ALL recognised coupling levels between federated clouds:

- A. Loosely coupled
- B. Partially coupled
- C. Tightly coupled
- D. Randomly coupled

Answers: A, B, C — Loosely coupled; Partially coupled; Tightly coupled

MSQ Q39. Motivations for cloud federation include:

- A. Maximize resource utilization
- B. Minimize power consumption
- C. Load balancing
- D. Expand a CSP's footprint
- E. Guarantee zero cost

Answers: A, B, C, D — Maximize resource utilization; Minimize power consumption; Load balancing; Expand a CSP's footprint

MSQ Q40. For high reliability, federated partners are ideally placed in different:

- A. Power zones
- B. Network zones
- C. Seismic zones
- D. The same rack

Answers: A, B, C — Power zones; Network zones; Seismic zones

Part D — VM Migration

MCQ Q41. In cold (non-live) migration, the source VM is:

- A. Kept fully running
- B. Turned off or suspended during migration
- C. Replicated infinitely
- D. Never moved

Answer: B. Turned off or suspended during migration

MCQ Q42. In hot (live) migration, the source VM:

- A. Is shut down first
- B. Continues to provide service during migration
- C. Loses all memory
- D. Cannot be resumed

Answer: B. Continues to provide service during migration

MCQ Q43. Which live-migration technique uses an iterative push phase followed by a stop-and-copy phase?

- A. Post-copy
- B. Pre-copy
- C. Cold copy
- D. Lazy copy

Answer: B. Pre-copy

MCQ Q44. Memory pages modified during pre-copy iterations are called:

- A. Clean pages
- B. Dirty pages
- C. Ghost pages
- D. Cache pages

Answer: B. Dirty pages

MCQ Q45. In post-copy migration, what is transferred FIRST?

- A. All memory pages
- B. The CPU/processor state
- C. The disk image
- D. The network stack

Answer: B. The CPU/processor state

MCQ Q46. Which hypervisors are stated to use the pre-copy technique?

- A. KVM, Xen, VMware
- B. Only VirtualBox
- C. Only Hyper-V
- D. None

Answer: A. KVM, Xen, VMware

MCQ Q47. For non-live migration of a single VM, migration time is approximately:

- A. $\text{VM size} \times R$
- B. $\text{VM size} / R$ (R = transmission rate)
- C. $R / \text{VM size}$
- D. Constant, independent of size

Answer: B. VM size / R (R = transmission rate)

MCQ Q48. In parallel migration of m VMs, each VM effectively receives what transmission rate?

- A. R
- B. $R \times m$
- C. R / m
- D. $2R$

Answer: C. R / m

MCQ Q49. 'Downtime' in migration refers to:

- A. Total time to move the VM
- B. Time the service remains unavailable to the user
- C. Time to boot the host
- D. Idle time of the CPU

Answer: B. Time the service remains unavailable to the user

MSQ Q50. Select ALL valid reasons to perform VM migration:

- A. Load balancing
- B. Server maintenance
- C. Power management (consolidation + sleep)
- D. Fault tolerance
- E. To increase hallucination

Answers: A, B, C, D — Load balancing; Server maintenance; Power management (consolidation + sleep); Fault tolerance

MSQ Q51. Which state/components must be handled when migrating a VM?

- A. CPU/processor state
- B. Main memory contents
- C. Network connectivity (IP/MAC/ARP)
- D. Active I/O devices
- E. The physical CPU chip itself

Answers: A, B, C, D — CPU/processor state; Main memory contents; Network connectivity (IP/MAC/ARP); Active I/O devices

MSQ Q52. Pre-copy iteration terminates when (select all):

- A. Number of rounds exceeds a threshold (n_{max})
- B. Total memory transmitted exceeds a threshold
- C. Number of dirty pages drops below a threshold
- D. The user logs out

Answers: A, B, C — Number of rounds exceeds a threshold (n_{max}); Total memory transmitted exceeds a threshold; Number of dirty pages drops below a threshold

MSQ Q53. Which are the phases/mechanisms of live migration mentioned?

- A. Push
- B. Stop-and-copy
- C. Pull (on-demand paging)
- D. Compile

Answers: A, B, C — Push; Stop-and-copy; Pull (on-demand paging)

Part E — Containers, Docker & Kubernetes

MCQ Q54. Containers are best described as:

- A. Hardware-level virtualization
- B. A lightweight OS-level virtualization technique
- C. A type of hypervisor
- D. A physical server

Answer: B. A lightweight OS-level virtualization technique

MCQ Q55. A key difference between containers and VMs is that containers:

- A. Virtualize at the hardware level
- B. Share the host OS kernel and virtualize at the OS level
- C. Each run a full guest OS
- D. Cannot be portable

Answer: B. Share the host OS kernel and virtualize at the OS level

MCQ Q56. Docker is described as an open platform to:

- A. Only store data
- B. Build, ship and run applications anywhere
- C. Manage electricity
- D. Cool data centers

Answer: B. Build, ship and run applications anywhere

MCQ Q57. Kubernetes is:

- A. A programming language
- B. An open-source system for automated deployment, scaling and management of containers
- C. A hypervisor
- D. A database

Answer: B. An open-source system for automated deployment, scaling and management of containers

MCQ Q58. In Kubernetes, the set of worker machines that run containerized applications are called:

- A. Pods
- B. Nodes
- C. Daemons
- D. Registries

Answer: B. Nodes

MCQ Q59. In Kubernetes, what manages the worker nodes and pods?

- A. The kubelet only
- B. The control plane
- C. The docker hub
- D. The BM25 index

Answer: B. The control plane

MCQ Q60. A Docker container is best described as:

- A. A static image on disk
- B. A runtime instance of an image
- C. A registry
- D. A hypervisor

Answer: B. A runtime instance of an image

MCQ Q61. The public registry that stores official Docker images is:

- A. Docker Hub
- B. GitHub
- C. CloudSim
- D. DynamoDB

Answer: A. Docker Hub

MCQ Q62. Docker images are built as:

- A. A single monolithic block
- B. Layered file systems
- C. Only binaries
- D. Raw disk dumps

Answer: B. Layered file systems

MCQ Q63. The Docker architecture's client and daemon communicate primarily over:

- A. FTP
- B. HTTP (REST API)
- C. SMTP
- D. Bluetooth

Answer: B. HTTP (REST API)

MCQ Q64. Which command saves a running Docker container as a new image?

- A. docker run
- B. docker commit
- C. docker pull
- D. docker tag

Answer: B. docker commit

MSQ Q65. Select ALL stated key benefits/features of Docker/containers:

- A. Separation of responsibilities
- B. Workload portability
- C. Application isolation
- D. Enables microservice architecture
- E. Requires reinstalling dependencies at every destination

Answers: A, B, C, D — Separation of responsibilities; Workload portability; Application isolation; Enables microservice architecture

MSQ Q66. Which are core Docker concepts/objects?

- A. Image
- B. Container
- C. Docker Hub (registry)
- D. Docker Engine
- E. Seismic zone

Answers: A, B, C, D — Image; Container; Docker Hub (registry); Docker Engine

MSQ Q67. Select ALL Docker commands mentioned in the lectures/demo:

- A. docker build
- B. docker run
- C. docker commit
- D. docker save / load
- E. docker migrate

Answers: A, B, C, D — docker build; docker run; docker commit; docker save / load

MSQ Q68. In the three deployment models, which statements are TRUE?

- A. Traditional: apps on physical servers with no resource boundary
- B. Virtualized: multiple VMs each with its own guest OS on a hypervisor
- C. Container: shares OS kernel, lightweight, portable
- D. Containers are heavier than VMs

Answers: A, B, C — Traditional: apps on physical servers with no resource boundary; Virtualized: multiple VMs each with its own guest OS on a hypervisor; Container: shares OS kernel, lightweight, portable

Part F — Dew Computing

MCQ Q69. In the natural analogy, dew computing sits at which level?

- A. Far in the sky
- B. Near the ground
- C. On the ground / at the device level
- D. In orbit

Answer: C. On the ground / at the device level

MCQ Q70. The two defining key features of dew computing are:

- A. Latency and jitter
- B. Independence and collaboration
- C. Encryption and pruning
- D. Scaling and billing

Answer: B. Independence and collaboration

MCQ Q71. Which is the classic real-world example of dew computing?

- A. Docker Hub
- B. Dropbox
- C. Kubernetes
- D. CloudSim

Answer: B. Dropbox

MCQ Q72. In dew computing, the local isolated environment that runs the dew server is called:

- A. A pod
- B. A Dew Virtual Machine (DVM)
- C. A hypervisor
- D. A container registry

Answer: B. A Dew Virtual Machine (DVM)

MCQ Q73. Dew computing is described as a ____ based paradigm that does not depend on a centralized server.

- A. Monolithic
- B. Micro-service
- C. Mainframe
- D. Hypervisor

Answer: B. Micro-service

MCQ Q74. 'Independence' in dew computing means the local device:

- A. Needs constant internet
- B. Can provide service WITHOUT continuous internet connectivity
- C. Cannot store data
- D. Must sync every second

Answer: B. Can provide service WITHOUT continuous internet connectivity

MCQ Q75. Which dew category matches GitHub-style development with synced settings/data?

- A. Storage in Dew
- B. Platform in Dew
- C. Web in Dew
- D. Database in Dew

Answer: B. Platform in Dew

MSQ Q76. Select ALL goals of the dew server:

- A. Data replication
- B. Data distribution
- C. Synchronization with the cloud
- D. Physical cooling

Answers: A, B, C — Data replication; Data distribution; Synchronization with the cloud

MSQ Q77. Select ALL categories/service models of dew computing mentioned:

- A. Web in Dew (WiD)
- B. Storage in Dew (SiD)
- C. Database in Dew (DiD)
- D. Software / Platform / Infrastructure in Dew
- E. Seismic in Dew

Answers: A, B, C, D — Web in Dew (WiD); Storage in Dew (SiD); Database in Dew (DiD); Software / Platform / Infrastructure in Dew

MSQ Q78. Which are challenges of dew computing?

- A. Power management
- B. Processor utility
- C. Data storage (local + sync)
- D. Information/database security
- E. Guaranteed zero latency

Answers: A, B, C, D — Power management; Processor utility; Data storage (local + sync); Information/database security

Part G — Serverless Computing

MCQ Q79. Serverless computing primarily shifts the focus from servers to:

- A. Networks
- B. Applications / programmers
- C. Cooling
- D. Storage disks

Answer: B. Applications / programmers

MCQ Q80. Whereas cloud computing targets system administrators, serverless mainly targets:

- A. Programmers/developers
- B. Electricians
- C. Auditors
- D. End customers only

Answer: A. Programmers/developers

MCQ Q81. The two broad categories of serverless technology are:

- A. IaaS and PaaS
- B. Function-as-a-Service (FaaS) and Backend-as-a-Service (BaaS)
- C. Hot and cold
- D. Push and pull

Answer: B. Function-as-a-Service (FaaS) and Backend-as-a-Service (BaaS)

MCQ Q82. Amazon's serverless FaaS platform is:

- A. Azure Functions
- B. Google Cloud Functions
- C. AWS Lambda
- D. Firebase

Answer: C. AWS Lambda

MCQ Q83. Microsoft's serverless function service is:

- A. Cloud Functions
- B. AWS Lambda
- C. Azure Functions
- D. Cosmos Lambda

Answer: C. Azure Functions

MCQ Q84. In serverless, you are billed:

- A. Per hour of server uptime
- B. Only when the function executes (per use)
- C. A flat monthly fee
- D. Per stored GB only

Answer: B. Only when the function executes (per use)

MCQ Q85. FaaS functions are typically described as:

- A. Long-running and stateful
- B. Event-driven, stateless and often ephemeral
- C. Always active
- D. GUI-based

Answer: B. Event-driven, stateless and often ephemeral

MCQ Q86. Backend-as-a-Service (BaaS) is used to replace:

- A. The client UI
- B. Server-side components (auth, database, storage)
- C. The CPU
- D. The network cable

Answer: B. Server-side components (auth, database, storage)

MCQ Q87. Google Firebase is given as an example of a:

- A. FaaS runtime
- B. Fully managed backend/database (BaaS)
- C. Hypervisor
- D. Load balancer

Answer: B. Fully managed backend/database (BaaS)

MCQ Q88. An AWS Lambda image size limit noted in the lecture is around:

- A. 5 MB
- B. 50 MB
- C. 500 MB
- D. 5 GB

Answer: B. 50 MB

MSQ Q89. Select ALL serverless FaaS providers named:

- A. AWS Lambda
- B. Google Cloud Functions
- C. Azure Functions
- D. OpenStack Nova

Answers: A, B, C — AWS Lambda; Google Cloud Functions; Azure Functions

MSQ Q90. Which are stated CHALLENGES/downsides of serverless computing?

- A. Complexity of asynchronous calls
- B. Shared-code breakage across functions
- C. Image size limits / cold-start warm-up time
- D. Too many tiny functions hurting maintainability
- E. Guaranteed lowest possible latency always

Answers: A, B, C, D — Complexity of asynchronous calls; Shared-code breakage across functions; Image size limits / cold-start warm-up time; Too many tiny functions hurting maintainability

MSQ Q91. Match providers to their databases (select all correct pairings):

- A. AWS → DynamoDB
- B. Google → Cloud Datastore/Firestore
- C. Azure → Cosmos DB
- D. AWS → Cosmos DB

Answers: A, B, C — AWS → DynamoDB; Google → Cloud Datastore/Firestore; Azure → Cosmos DB

Part H — Sustainable Cloud Computing

MCQ Q92. The primary goal of sustainable cloud computing is to minimize energy consumption and carbon footprint while ensuring:

- A. Maximum hardware
- B. Reliability of the data centers
- C. Zero users
- D. Maximum latency

Answer: B. Reliability of the data centers

MCQ Q93. Which component is stated to consume the largest share (~45%) of a server's energy?

- A. Cooling
- B. Processor
- C. Memory
- D. Networking

Answer: B. Processor

MCQ Q94. Approximately what share of energy is attributed to cooling?

- A. 5%
- B. 20%
- C. 45%
- D. 70%

Answer: B. 20%

MCQ Q95. DVFS stands for:

- A. Dynamic Virtual File System
- B. Dynamic Voltage and Frequency Scaling
- C. Distributed Verified Fog Service
- D. Data Volume Flow Sensor

Answer: B. Dynamic Voltage and Frequency Scaling

MCQ Q96. Projected cloud data-center energy consumption is expected to reach roughly what by 2030?

- A. 800 TWh
- B. 8,000 TWh (8k TWh)
- C. 80 TWh
- D. 8 TWh

Answer: B. 8,000 TWh (8k TWh)

MCQ Q97. PUE, often targeted in sustainability, stands for:

- A. Power Usage Effectiveness
- B. Peak Utilization Energy
- C. Public Utility Estimate
- D. Processor Unit Efficiency

Answer: A. Power Usage Effectiveness

MCQ Q98. Which technique reuses waste heat while cooling the data center?

- A. DVFS
- B. Vapor-absorption-based cooling
- C. Magnitude pruning
- D. Negative sampling

Answer: B. Vapor-absorption-based cooling

MCQ Q99. In the conceptual model, which module decides between renewable and conventional (grid) energy?

- A. Cooling manager
- B. Power manager
- C. Remote CDC manager
- D. Client module

Answer: B. Power manager

MCQ Q100. An Automatic Transfer Switch (ATS) is used to:

- A. Cool the CPU
- B. Manage/switch energy coming from both (renewable + conventional) sources
- C. Migrate VMs
- D. Store containers

Answer: B. Manage/switch energy coming from both (renewable + conventional) sources

MSQ Q101. Select ALL renewable energy sources mentioned for CDCs:

- A. Solar
- B. Wind
- C. Hydro (water)
- D. Coal

Answers: A, B, C — Solar; Wind; Hydro (water)

MSQ Q102. Select ALL modules of the sustainable-cloud conceptual model:

- A. Cloud architecture (IaaS/PaaS/SaaS)
- B. Cooling manager
- C. Power manager
- D. Remote CDCs
- E. Marketing manager

Answers: A, B, C, D — Cloud architecture (IaaS/PaaS/SaaS); Cooling manager; Power manager; Remote CDCs

MSQ Q103. Which are recognised aspects/components of achieving sustainable cloud computing?

- A. Thermal-aware scheduling
- B. Renewable energy management
- C. Waste-heat utilization
- D. Capacity planning / predictive models
- E. Increasing carbon footprint

Answers: A, B, C, D — Thermal-aware scheduling; Renewable energy management; Waste-heat utilization; Capacity planning / predictive models

MSQ Q104. Which trade-off must sustainable cloud computing balance?

- A. Energy/carbon reduction vs SLA reliability
- B. Replication for reliability increases energy demand
- C. Turning servers on/off affects reliability
- D. More energy always improves user experience

Answers: A, B, C — Energy/carbon reduction vs SLA reliability; Replication for reliability increases energy demand; Turning servers on/off affects reliability